CHAMBOLI SECONDARY SCHOOL BOARD

LESSON PLAN

DEPARTMENT: DESIGN AND TECHNOLOGY **DATE:** ……2023.

TEACHER: KHONDOWE J **DURATION**: 120 MINUTES

SUBJECT: DESIGN AND TECHNOLOGY **NO. OF PUPILS:**

TOPIC: DESIGN FOLIO **CLASS:**

OBJECTIVES: At the end of the lesson, PSBAT:

1. Define Design process?
2. Define Design Folio?
3. Explain the Theme

TEACHING AIDS/REFERENCES: Chalk board and chalk, Design and technology pupils' book.

PART	TIME	LESSON CONTENT	METHODOLOGY	LEARNER ACTIVITY
INTRODUCTION	5 MIN	Design process is a step by step process of product design, with series of activities at different design stages which will help the artisan in producing a solution in form of product to address an identified problem. The design process has a very important place in Craft, Design and Technology. It is sometimes called product design. Important aspects to consider in the activities include; material, Construction, size, safety, Function various ideas to mention but a few. This information must be well presented and well laid out so that people can look at and understand.	✓ Verbal Exposition	
DEVELOPM	95 MIN	➢ . Design Folio-this term is derived from the collection of the leaners performances to show their work and accomplishment over time. ➢ The Theme- A theme is a topic or word that can be analyzed. A theme should have a wide range of characteristics that gives a designer room to generate interest and	✓ Verbal Exposition	Writing and answering questions

E N T		consider what can be made without tracking off from the area of interest. ▪ Examples of themes that can be analyzed are shown below; SAFETY RECREATION LEISURE LITERACY STORAGE HOUSEHOLD These words are usually nouns that can be analyzed and learners can come up with different designs that may vary in Size and shape. As an example we can analyze the word SAFETY which we have chosen from the themes that are listed above. The first page therefore will exclusively show the theme. At this stage, the pages don't have to be numbered because the table of content page will be inserted in after the work is done.	✓ Question and Answer ✓ Discussion	
W R A P U P	20 MIN	Task 1. Design and make a folder where you will keep your work. 2. Other than the theme SAFETY we have adopted in this module, Design a theme page from any one of the given themes.	VERBAL EXPOSION	WRITING HOME WORK IN THEIR EXERCISE BOOKS

LESSON CRITIQUE:

……

……

……

……

CHAMBOLI SECONDARY SCHOOL BOARD

LESSON PLAN

DEPARTMENT: DESIGN AND TECHNOLOGY **DATE: :** ……2023.

TEACHER: KHONDOWE J **DURATION**: 120 MINUTES

SUBJECT: DESIGN AND TECHNOLOGY **NO. OF PUPILS:**

TOPIC: DESIGN FOLIO **CLASS:** …….

OBJECTIVES: At the end of the lesson, PSBAT:

1. Explain the Theme Definitions
2. Define and explain what situation is
3. Explain the three main parts of a situation

TEACHING AIDS/REFERENCES: Chalk board and chalk, Design and technology pupils' book.

PART	TIME	LESSON CONTENT	METHODOLOGY	LEARNER ACTIVITY
INTRODUCTION	5 MIN	Recap of the previous lesson Design process is a step by step process of product design, with series of activities at different design stages which will help the artisan in producing a solution in form of product to address an identified problem. ➢ . Design Folio-this term is derived from the collection of the leaners performances to show their work and accomplishment over time. ➢ The Theme- A theme is a topic or word that can be analyzed	✓ Verbal Exposition	
DEVELOPM	95 MIN	The definitions can be obtained from various sources such as dictionaries, encyclopedias, magazines, journals, individuals and one of them must be of your own opinion or understanding. A minimum of four definitions will be required for you to demonstrate that you have a correct understating of the theme. Definitions must be clearly referenced. Always acknowledge	✓ Verbal Exposition	Writing and answering questions

E N T		the source of your information. Def one: The condition of being safe. Ref :the concise oxford English Dictionary (eleventh Edition) Def two: A state in which or a place where you are safe and not in danger or at risk. Ref : Cambridge advanced learners Dictionary (Third Edition) Def Three: A device designed to prevent injury or accident. Ref: Advanced English Dictionary (Mobile Version) Working Definition: Creating and upholding measures that reduce risk on people and their environment. We are required to give references that meet ABC referencing system where they will be required to state the reference as follows: Editors(Authors, Surname, initials), Book Title, Year of publication, Place of publication and publisher.	✓ Question and Answer ✓ Discussion	
W R A P U P	20 MIN	Task 1. Write three Definitions from different sources and clearly show the references. 2. Write a working definition that is derived from the definitions.	VERBAL EXPOSION	WRITING HOME WORK IN THEIR EXERCISE BOOKS

LESSON CRITIQUE:

……

……

……

……

CHAMBOLI SECONDARY SCHOOL BOARD

LESSON PLAN

DEPARTMENT: DESIGN AND TECHNOLOGY **DATE:**…… 2023.

TEACHER: KHONDOWE J **DURATION**: 120 MINUTES

SUBJECT: DESIGN AND TECHNOLOGY **NO. OF PUPILS:**

TOPIC: DESIGN FOLIO **CLASS:** ……

OBJECTIVES: At the end of the lesson, PSBAT:

1. Define and explain what situation is
2. Explain the three main parts of a situation

TEACHING AIDS/REFERENCES: Chalk board and chalk, Design and technology pupils' book.

PART	TIME	LESSON CONTENT	METHODOLOGY	LEARNER ACTIVITY
INTRODUCTION	5 MIN	Recap of the previous lesson. There are areas where you need to consider what specific area you might need to concentrate on so that you can precisely write the design brief it is simply a narrow down of the theme that we have been analyzing so that we can have a situation that is linked to the theme.	✓ Verbal Exposition	
DEVELOPMEN	95 MIN	SITUATION:- A situation is a statement or picture that depicts the condition in which people live in, or a state in which things are found or seen. For easy analysis, a situation can be broken in to three parts: Example: My younger brother, who is aged three years, falls from his bed at times when he is asleep in his bedroom, and usually he sustains injures. 1. Activity –The activity is sleeping 2. Concern- The concern is that the boy falls off from his bed 3. Place- The place is Bedroom	✓ Verbal Exposition ✓ Question and Answer	Writing and answering questions

T		When you look closely at this situation it pays attention to domestic accidents and particular attention is paid to a three year boy. PROBLEM IDENTIFICATION:- A problem is always derived from a situation. Failure to correctly identify the problem may lead to failure to address the condition that affects the people. There are possibilities that more than one problem can be identified in a given situation. Our task would be to address ourselves to a particular problem even if they are many problems that might be identified, so great caution must be exercised so that we don't stray from our key area of concentration. E.G The bed is not secure enough to prevent the child falling from the bed.	✓ Discussion	
W R A P U P	20 MIN	Task 1. Write five situations 2. Identify a problem from the situation you have drawn.	VERBAL EXPOSION	WRITING HOME WORK IN THEIR EXERCISE BOOKS

LESSON CRITIQUE:

……
……
……
……

CHAMBOLI SECONDARY SCHOOL BOARD

LESSON PLAN

DEPARTMENT: DESIGN AND TECHNOLOGY **DATE:** ……2023.

TEACHER: KHONDOWE J **DURATION**: 120 MINUTES

SUBJECT: DESIGN AND TECHNOLOGY **NO. OF PUPILS:**

TOPIC: DESIGN FOLIO **CLASS:** ……

OBJECTIVES: At the end of the lesson, PSBAT:

1. Define and explain Design Brief
2. Explain the three main parts of a situation
3. Example of a design Brief
4. Define and analyze Research.

TEACHING AIDS/REFERENCES: Chalk board and chalk, Design and technology pupils' book.

PART	TIME	LESSON CONTENT	METHODOLOGY	LEARNER ACTIVITY
INTRODUCTION	5 MIN	Recap of the previous lesson.	✓ Verbal Exposition	
DEVELOPMEN	95 MIN	DESIGN BRIEF A design brief is defined as a clear and short statement that clearly shows what you intend to do. A good design brief should not specify names of products to be made such as design and make a bed. Having mentioned a bed, it limits the need to make any attempt to design anything else since a bed has already been proposed to be made. The brief like in the situation should include three components 1. User 2. Function 3. Place.	✓ Verbal Exposition ✓ Question and	Writing and answering questions

T		Example of design Brief Design and make an artifact that will prevent my brother from falling when he is asleep in bed in his bedroom. NOTE: If you are required to only design the product so that other people can make it, then your design brief will read, Design an artifact that will prevent my brother from falling when he is asleep in bed in his bedroom. RESEACH The word research simply means a systematic investigation to establish facts. After research has been completed recommendations that will lead to the formulation of specifications will be made Methods of research 1. Questionnaire 2. Interviews 3. Desk/internet 4. observation	Answer ✓ Discussion	
W R A P U P	20 MIN	Task 1. Define and explain Design Brief 2. Explain the three main parts of a situation 3. Example of a design Brief 4. Define and analyze Research	VERBAL EXPOSION	WRITING AN EXERCISE

LESSON CRITIQUE:

……

……

……

……

CHAMBOLI SECONDARY SCHOOL BOARD

LESSON PLAN

DEPARTMENT: DESIGN AND TECHNOLOGY **DATE:** ……2023.

TEACHER: KHONDOWE J **DURATION**: 120 MINUTES

SUBJECT: DESIGN AND TECHNOLOGY **NO. OF PUPILS:**

TOPIC: Structures **CLASS:** ……

OBJECTIVES: At the end of the lesson, PSBAT:

1. Define structures
2. Explain types of structures
3. Give Examples of structure.

TEACHING AIDS/REFERENCES: Chalk board and chalk, Design and technology pupils' book.

PART	TIME	LESSON CONTENT	METHODOLOGY	LEARNER ACTIVITY
INTRODUCTION	5 MIN	All around we are surrounded by different structures. These structures are divided into groups according to their origin. In this system, structures are divided into natural and man made structures.	✓ Verbal Exposition	
DEVELOP	95 MIN	<u>Natural structures</u> Are not made by people such as eggshells, spider webs, caves and trees. Examine a birds feather which is a natural structure. It has a definite shape and has a pattern. Feathers have many functions on a bird	✓ Verbal Exposition	Writing and answering questions

M E N T		➢ They protect it from rain ➢ Allow a bird to fly ➢ Insulate a bird in cold. <u>Man made structures</u> Many things built by people are man-made structures. For example houses, bridges, shopping bags, footballs, cups, and spoons are all manufactured structures. Many man-made structures are modelled after natural structures. For example a fishing net is designed similar to spider web. Structures are divided into three groups;- 1. Frame 2. Shell 3. Mass or solid structures	✓ Question and Answer ✓ Discussion	
W R A P U P	20 MIN	Task 1. Name other natural, non-living structures around your school. 2. Name other man-made structures similar to natural structure around your school.	VERBAL EXPOSION	WRITING AN EXERCISE

LESSON CRITIQUE:

……

……

……

……

LESSON PLAN

DEPARTMENT: DESIGN AND TECHNOLOGY **DATE:** ……2023.

TEACHER: KHONDOWE J **DURATION**: 120 MINUTES

SUBJECT: DESIGN AND TECHNOLOGY TOTAL **NO. OF PUPILS:…….**

TOPIC: Adhesive **BOYS…….. GIRLS ….. CLASS:** ……..

Out comes: At the end of the lesson, PSBAT:

1. Classify Adhesives
2. Choose the correct adhesive for the job
3. Apply correct gluing techniques
4. State advantages and disadvantages of glue
5. List abbreviations of some adhesives

TEACHING AIDS/REFERENCES: Chalk board and chalk, Design and technology pupils' book.

PART	TIME	LESSON CONTENT	METHODOLOGY	LEARNER ACTIVITY
INTRODUCTION	5 MIN	Adhesives are bonding substances used for fixing joints **permanently**, Adhesives are commonly called glues which can be grouped into three varieties:- 1. Protein adhesives 2. Synthetic resin adhesive 3. Contact adhesives The choice of these adhesives depends on their individual properties.	✓ Verbal Exposition	
DEVELO	95 MIN	1. Classify Adhesives I. Protein adhesives These are animal glue, Casine glue, vegetable glue and blood albumin glue. II. Synthetic resin adhesive These are urea formaldehyde resin (U.F), melamine formaldehyde (M.F), Phenol	✓ Verbal Exposition	Question and answer

P M E N T		formaldehyde (P.F), Resorcinol phenol-formaldehyde, epoxy and polyvinal acetate (P.V.A) III. Contact adhesives These are instant bonding, they set and dry by evaporation. It may be allowed to set between 10-20 minutes. They are mostly used to for bonding dissimilar materials such as Formica, fabrics, leather, glass, and plywood and particle boards. 1. Choose the correct adhesive for the job 2. Apply correct gluing techniques 3. State advantages and disadvantages of glue 4. List abbreviations of some adhesives	✓ Question and Answer ✓ Discussion	
W R A P U P	20 MIN	Task 1. Classify Adhesives 2. Choose the correct adhesive for the job 3. Apply correct gluing techniques 4. State advantages and disadvantages of glue 5. List abbreviations of some adhesives	VERBAL EXPOSION	WRITING AN EXERCISE

LESSON CRITIQUE:

……

……

……

……